

Multiple Choice

1. **Answer:** A

Options: A) Living will; B) DNH; C) DNR; D) Durable power of attorney

Note: A living will is a legal document that allows individuals to outline their preferences for medical treatment, specifically detailing the types of interventions they wish to receive or avoid in the event that they are unable to communicate their wishes.

2. **Answer:** A

Options: A) Care for an ailing parent while continuing to work; B) Work a high stress job while their children are at home; C) Pay for nursing home care by sacrificing their own care; D) Serving as an executor of their parent's estate while still working full time

Note: The term "work-family conflict" accurately describes the situation where middle-aged adults experience competing demands from their job and the responsibility of caring for an ailing parent, leading to stress and potential role strain.

3. **Answer:** C

Options: A) A sharp and rapid epidemic involving more than one country; B) An outbreak which recurs again and again; C) A rapid global outbreak starting from a single focus; D) A characteristic of common cold virus and HIV

Note: A virus pandemic is characterized by the widespread and rapid transmission of a virus across multiple countries or continents, originating from a single point of infection, which highlights the global nature and significant impact of such outbreaks on public health.

4. **Answer:** A

Options: A) Transfusion blood is screened; B) Vaccine has been developed; C) A range of new antivirals; D) Social distancing

Note: Transfusion blood is screened to ensure that it is free from parvoviruses, thereby reducing the risk of transmission and protecting patients from potential viral infections associated with blood transfusions.

5. **Answer:** D

Options: A) Change in behavior; B) Comprehensive coverage; C) Effective use of behavioral theory; D) Sustained behavior change

Note: The most important goal of a behavioral intervention is sustained behavior change because it ensures that the positive modifications in behavior are maintained over time, leading to long-term health benefits and improved quality of life.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Impaired renal function is not commonly associated with bunyavirus infections, as these viruses primarily cause febrile illnesses and hemorrhagic fevers rather than renal complications.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because while certain injuries can severely impact respiratory function, many patients are still capable of breathing independently, albeit with difficulty or reduced capacity, without immediate reliance on life support.

8. **Answer:** True

Options: A) True; B) False

Note: The phase of disorganization and despair often encompasses a prolonged period of emotional turmoil and adjustment, as individuals grapple with feelings of loss and the challenge of reestablishing their lives after a significant bereavement.

9. **Answer:** False

Options: A) True; B) False

Note: Excess disability in Alzheimer's disease can manifest at various stages of the illness, not just in the last stage, as cognitive and functional impairments can occur throughout the progression of the disease.

10. **Answer:** False

Options: A) True; B) False

Note: Blood samples are crucial for diagnosing hepatitis B infections as they allow for the detection of specific hepatitis B antigens and antibodies, which indicate the presence of the virus in the bloodstream.

Long Form Response

11. **Answer:** Virus nucleocapsids are primarily structured in two geometric forms: icosahedral and helical. Icosahedral nucleocapsids, characterized by their spherical symmetry and composed of 20 triangular faces, are efficient in enclosing genetic material with minimal protein, which allows for stability and protection. In contrast, helical nucleocapsids are elongated and cylindrical, with the viral RNA or DNA wrapped in a spiral manner around the protein coat, providing flexibility and ease of assembly. These structural differences have significant implications for viral function; for instance, icosahedral viruses tend to be more resilient to environmental stresses, while helical viruses often exhibit a simpler assembly process that can facilitate rapid infection. Understanding these geometric structures aids in comprehending how viruses infect host cells and the strategies they employ to evade the immune system.

Note: The answer correctly identifies and contrasts the structural characteristics of

icosahedral and helical nucleocapsids, explaining how these geometric forms influence the efficiency of genetic material encapsulation and the implications for viral stability, protection, and assembly during infection.

12. **Answer:** The vagus nerves play a critical role in the functioning of the oesophagus as it passes through the diaphragm, primarily by regulating the autonomic nervous system's control over digestive processes. Anatomically, the vagus nerves provide parasympathetic innervation to the oesophagus, influencing peristalsis and the secretion of digestive enzymes. This innervation is essential for coordinated swallowing and the effective movement of food toward the stomach. Health implications arise when the vagus nerve is compromised- conditions such as gastroesophageal reflux disease (GERD) or esophageal motility disorders can result from dysfunction in this pathway, highlighting the importance of the vagus nerves in maintaining oesophageal health and overall digestive function. Understanding their anatomical significance is crucial for diagnosing and treating related health issues.

Note: correct because it accurately describes the vagus nerves' anatomical role in providing parasympathetic innervation to the oesophagus, which is essential for regulating peristalsis and digestive enzyme secretion, thereby influencing overall digestive health.